

UCSC Extension Course: AISV.X401.(21) Summer 2025

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Project Abstract

1. Problem Statement

Businesses collect a lot of customer feedback through surveys and reviews, but it takes a lot of time and effort to read and understand all these comments. The goal of this project is to create a computer program that can automatically read customer feedback and tell if it is positive, negative, or neutral. This will help companies quickly understand what customers like or dislike and improve their products and services.

2. Dataset

I will use a [Customer Feedback Dataset](#) from Kaggle that contains many customer comments with labels showing whether the feedback is positive, negative, or neutral. This dataset will help train and test the program to recognize the sentiment in new feedback.

3. Proposed Techniques to be Used

- I will use a modern language processing tool called BERT, which is a computer model designed to understand human language.
- The model will be trained to read customer comments and classify their sentiment.
- I will also use simple methods to clean and prepare the text data before training the model.
- After training, I will check how well the model works by comparing its predictions to the actual labels.

4. Steps to Complete the Project

1. Look at the dataset and clean the text data to remove any errors or unnecessary parts.
2. Prepare the data so it can be used by the BERT model.

3. Train the BERT model to classify the sentiment of customer comments.
4. Test the model to see how accurate it is.
5. Analyze which words or phrases the model uses to make its decisions.
6. Summarize the results and explain how this tool can help businesses understand customer feedback faster.

5. Expected Output/Results

- A working model that can automatically tell if customer feedback is positive, negative, or neutral.
- A report showing how accurate the model is and examples of its predictions.
- Visual charts that show the overall sentiment trends in the feedback.
- Insights on how this model can help companies improve their customer service and products.